

Cryptography

by - manu sir

Types of Encryption Algorithms

Symmetric algorithms:

- Sender and Receiver use two instances of the same key for encryption and decryption.
- Key has dual functionality-it can carry out both encryption and decryption process.

Advantages:

- Much faster than asymmetric systems.
- Hard to break if using a large key size.

Disadvantages:

- Scalability and key Management
 - Equation = $N(N-1)/2$ = number of keys
- Key exchange.
- Authentication and Non-repudiation.

One scenario:

- Prime number 'p' agreed by both Tinku and Pinky: 23
- Base 'g' agreed by both Tinku and Pinky: 5
- Secret integer selected by Tinku (a): 6
- Secret Integer selected by Pinky (b): 15
- Tinku calculates 'A = $g^a \text{ mod } p$ ', which is $5^6 \text{ mod } 23 = 8$
- Pinky calculates 'B = $g^b \text{ mod } p$ ', which is $5^{15} \text{ mod } 23 = 19$
- Tinku sends 'A' to Pinky
- Pinky send 'B' to Tinku
- Tinku calculates the secret key 's = $B^a \text{ mod } p$ ', which is : 2
- Pinky calculates the key secret key 's = $A^b \text{ mod } p$ ', which is : 2

Example:

- Data encryption Standard (DES)
- Triple-DES (3DES)
- Blowfish
- Twofish
- IDEA
- RC4, RC5, RC6

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- Advanced Encryption Standard (AES)

Types of symmetric algorithm:

- Block Cipher
- Stream Cipher

Block cipher:

Block Cipher standards:

- Data encryption Standard
- 2DES
- Triple DES
- Advanced Encryption Cipher

Block Cipher- Algorithms

- Blowfish
- Twofish

International Data Encryption Algorithm (IDEA)

- RC2
- RC5
- RC6

Stream Cipher

- Stream Cipher-Algorithms
 - RC4

Note: Get the key size of all the algorithm which is covered in class for exam.

Asymmetric Algorithms:

- In this algorithm, every user will have two keys.
 - Public key.
 - Private key.
- Every user will have their own unique public key and private key.
- Public key can be distributed to anyone, whereas private is always kept confidential.
- Encryption using Public key provides Authentication, where Private key provides Confidentiality.

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- If a message is encrypted using A's public key can only be decrypted by the A's private key.
- If a message is encrypted by using A's private key can only be decrypted by the A's public key.
- If a communication is going on between A and B then in that case, B will be having A's Public key and A will be having B's public key.
- So, if they want to establish a secure communication then they will encrypt the message using receiver's public key. And receiver will decrypt the message using their private key.
- If the communication is meant for authentication and there is no secrecy user wants to keep, then user will encrypt the message using their private key, where everyone decrypts the message using sender's public key. So, they will authenticate that this message is came from the that sender from which they are expecting.

Advantages:

- Better key distribution than symmetric systems.
- Better Scalability than symmetric systems.
- Can provide authentication and non-repudiation.

Disadvantages:

- Slower than symmetric.

Example:

- RSA
- DSA
- Elliptic curve cryptosystem (ECC)
- Diffie-Hellman
- EL-Gamal
- Digital Signature Algorithm (DSA)
- Knapsack

Cryptographic Hash Function:

- Function that takes a variable-length string as a message and produces a fixed-length value called a hash value.

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- A mathematical algorithm that maps data of arbitrary size to a bit string of a fix size.
- One-way function.
- In linux to convert the hash value we use the following command;

```
#md5sum one.txt
```

Asymmetric encryption Lab:

On machine 1:

1. # sudo apt-get install rng-tools
2. # mkdir /home/user1/demo
3. # cd /home/user1/demo
4. # mkdir files keys
5. # cd keys
6. # mkdir private public
7. # sudo apt-get install tree
8. # tree -----(to check that the directories)
9. If the rng tools did not start properly do the following steps otherwise move to the step 10.
 - a. # sudo nano /etc/default/rng-tools
 - b. Add line → HRNGDEVICE=/dev/urandom
 - c. Save the file
 - d. # sudo service rng-tools start or # sudo systemctl restart rng-tools
10. # gpg --gen-key
 - a. Enter the "real name: <any name>"
 - b. Enter the "Email: <any mail>"
 - c. Enter "O" to continue
 - d. Enter the paraphrase
11. # gpg --list-keys -----(to see the public key)
12. # gpg --list-secret-keys -----(to see the private key)
13. Go to the machine 2
14. After getting back from the machine 2 you should have machine 2's public key in public folder.
15. Then we have to import that key in machine 1
16. # gpg --import /home/user1/demo/keys/public/public-gpg-key.asc
17. # gpg --list-key -----(user 2's public should be listed)
18. # touch /demo/file.txt
19. # echo abcd > file.txt
20. # gpg --output file.txt.gpg --recipient <key> --encrypt file.txt

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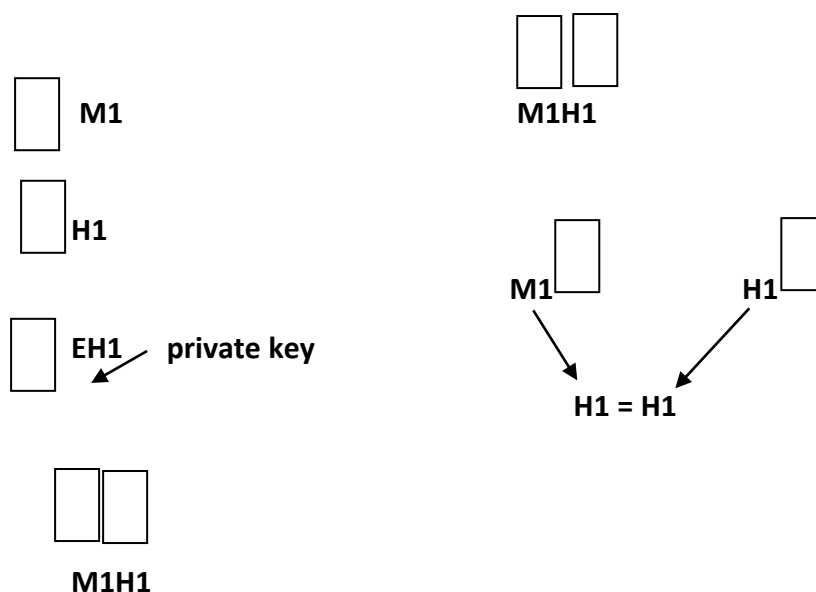
21. Copy the encrypted file which is .gpg file to machine 2
22. # scp /home/user1/demo/files/file.txt.gpg
user2@<IP>:/home/user2/demo/files/
23. Go the machine 2 on step 7

On machine 2:

Do same as on the machine 1 from step 1 to 11:

1. # gpg --list-keys
2. # gpg --export --armor <public_key>>/home/machine2/demo/keys/public
/public-gpg-key.asc
3. # tree
4. After exporting we need to copy the public-gpg-key.asc to the machine 1
5. Scp /home/user2/demo/keys/public/public-gpg-key.asc
user1@<IP>:/home/user1/demo/keys/public/
6. Go to the machine 1 on step 14
7. # gpg --decrypt files/file.txt.gpg
8. # gpg --output <newfile>.txt --decrypt file/file1.txt ---(in case you want to
decrypt the file and save the output in some other file)

Digital Signature:



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Machine1:

```
#gpg --clearsign <file>.txt
```

Machine2:

```
#gpg --verify <file>.txt.gpg
```

Encrypt and sign:

Symmetric

```
# gpg --sign --symmetric file.txt
```

```
# gpg --decrypt file.txt.gpg
```

Asymmetric

```
# gpg -s -e -r user2@shuhari.local file1.txt
```

```
# gpg --decrypt file1.txt
```

GPG:

Symmetric:

Encryption / Decryption – ASCII

Encryption / Decryption – Binary

Asymmetric:

Encryption / Decryption – ASCII

Encryption / Decryption – Binary

Hashing (using MD5):

Md5sum (linux)

Hashcalc (Windows)

Symmetric / Asymmetric Concepts (Theory)